

Erik Bloomquist

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EDUCATION

The Ohio State University, Columbus, OH 2026 - Present
Doctor of Philosophy (Ph.D.), Electrical & Computer Engineering
GPA: 4.00/4
Advisor: Dr. Zak Kassas

University of Florida, Gainesville, FL Aug. 2022 - May 2026
Bachelor of Science, Electrical Engineering (BSEE)
GPA: 3.98/4
Honors: Summa Cum Laude
Undergraduate Honors Thesis: Spectral Feature Extraction for Reinforcement Learning in Financial Markets Under Nonstationary Dynamics
Advisor: Dr. Catia Silva

WORK EXPERIENCE

Undergraduate Teaching Assistant | *UF ECE Dept.* Jan. 2026 - May 2026

- Assisted Dr. Catia Silva in operation of "Applied Machine Learning Systems" course
- Course topics included data processing and experimental design, implementation of various regression, classification, clustering, and dimensionality reduction algorithms with Scikit-Learn, and implementation of various deep learning architectures (MLPs, CNNs, Autoencoders, RNNs, and Transformers) using Pytorch

Undergraduate Teaching Assistant | *UF ECE Dept.* Aug. 2024 - Dec. 2025

- Assisted Dr. Catia Silva in operation of "Fundamentals of Machine Learning" course
- Course topics included Bayesian learning, generative models, clustering, discriminative classification, dimensionality reduction, manifold learning, neural networks, and deep learning

Undergraduate Teaching Assistant | *UF Engineering Education Dept.* May 2023 - July 2024

- Assisted Dr. Lilianny Virguez in operation of "Elements of Electrical Engineering" course
- Course topics included AC & DC circuit analysis, Arduino projects, and basic C++ programming

Student Assistant- Database | *UF College of Dentistry* May 2023 - Sept. 2023

- Collaborated with the UF College of Dentistry to migrate 5000 medical records into the College's database

RESEARCH EXPERIENCE

Spectral Feature Extraction for Reinforcement Learning in Financial Markets | *Dr. Catia Silva*

- Designed a reinforcement learning framework for financial market prediction using spectral features; served as project leader and sole student author
- Work supported by grant funding from UF Center for Undergraduate Research's AI Scholars Program

Literature Review: Adversarial Machine Learning | *Dr. Damon Woodard*

- Surveying adversarial ML attacks and defenses to assess effectiveness and guide future research in robust deep learning
- Work supported by the Florida Institute for National Security (FINS)

From Black-Box to Glass-Box Models: Towards Explainability in Reinforcement Learning | *Dr. Damon Woodard*

- Trained common RL algorithms (DQN, PPO) on benchmark environments (Cart Pole, Mountain Car, Lunar Lander)
- Examined decision boundaries and extracted interpretable policy representations

AWARDS

Electric E Award | *Recognition* May 2026

- Granted the highest honor from the UF ECE Department for graduating with a GPA above 3.9

Dr. Joseph S. Rosko Award | *Scholarship* Aug. 2025

- Awarded \$5000 scholarship for academic achievement and involvement in the UF ECE department

AI Scholars Program | *Research Funding* May 2025

- Awarded \$1750 grant to lead independent research bridging machine learning and financial markets

Florida Institute for National Security (FINS) Talent Pipeline | *Fellowship* Feb. 2024

- Awarded \$5000 fellowship to perform research at the intersection of AI and national security

PRESENTATIONS

2026 Spring Undergraduate Research Symposium <i>Poster Presentation</i>	Apr. 2026
– Presented research in reinforcement learning for financial market prediction using spectral information	
2025 Spring Undergraduate Research Symposium <i>Poster Presentation</i>	Apr. 2025
– Presented work on interpretable reinforcement learning	

LEADERSHIP & INVOLVEMENT

Undergraduate Curriculum Chair <i>Eta Kappa Nu (HKN) Epsilon Sigma Chapter</i>	Jan. 2024 - May 2026
– Represented IEEE’s honor society at faculty curriculum committee meetings	
– Selected to serve on the ECE student panel, answering curriculum-related questions from underclassmen in electrical and computer engineering	
– Spoke at the IEEE ”Find Your Field” event, helping expose underclassmen ECE students to the machine learning field	

PROJECTS

Spectral Feature-Based Stock Trading Agent
– Designed and implemented a reinforcement learning framework for financial market prediction using spectral features. Documented the impact of spectral penalty terms in the reward function on agent performance.
ECG Anomaly Detection Models
– Developed several machine learning pipelines for real-time detection of cardiac anomalies using the MIT-BIH Arrhythmia Database. Compared classical ML algorithms and deep learning models, with emphasis on feature extraction, signal preprocessing, and classification performance.
Autonomous Tabletop Poker-Playing System (CardShark)
– Designed and created an embedded, computer-vision–driven platform built around a Raspberry Pi that captures images of the table, identifies cards and game state, and uses decision logic to determine optimal actions. It integrates custom hardware (controllers, cameras, and analog circuitry) with software for image processing and strategy, enabling semi-autonomous or autonomous play in a physical poker environment.
Blackjack Game System
– Designed and implemented an embedded blackjack game using a Raspberry Pi Pico, integrating an I2C character LCD, PIR sensor, push-button inputs, and DAC-driven audio output. Built custom PCB with regulated power rails, signal conditioning, and an LM386-based amplifier.

SKILLS

Programming & Tools: Python (PyTorch, TensorFlow, scikit-learn, NumPy, Pandas), MATLAB, C++
Data & Tools: Git, Altium Designer, LTspice, LaTeX, Jupyter, SQL
Machine Learning: Deep Learning, Reinforcement Learning, Adversarial ML, End-to-End ML Systems, Data Cleaning & Preprocessing
Signals & Systems: Digital Signal Processing, Biomedical Signal Analysis, Circuit Modeling
Professional Skills: Technical Writing, Effective Communication, Poster Presentations, Teaching & Mentorship

RELEVANT COURSEWORK

Machine Learning Mathematics	Applied Machine Learning Systems
Formal Methods for Robotics & AI	Data Science
Stochastic Methods	Communication Systems
Digital Signal Processing	Biomedical Image Analysis
Analog Circuitry	Electrophysics